Alex Hale's Board Map Legend

Oct 5, 2026 · @Alex

Quick start

Alex Hale's Board (called Joseph's Board until Oct 7, 2026; the folder and file names still say josephs) is a market dashboard plus two trading robots ("algos"), all running 24/7 on the spare Mac (Alexs-MBP) in the folder `~/josephs-board`. You control it from Terminal with the `./jb` command, and you look at it in a web browser.

The five things you'll use most:

1. `cd ~/josephs-board` first, every time you open Terminal.
2. `./jb status` to see if every part is running.
3. `./jb restart` to fix most problems. It is safe and loses nothing.
4. On your phone, open **http://100.93.57.24:8090/** (Tailscale must be on).
5. A green dot on a board tab means that part is running.

Where this legend came from: the code you uploaded (bridges, web, IBKR algo, lib, gateway), the two fixes installed on Oct 5 (delayed prices and the Webull fix), and your chat notes. Anything marked *(from notes)* came only from your notes, because that code wasn't in the upload. Every board tab also has a "? Legend" button that explains that tab on the spot, once the legend update is installed.

./jb commands

Type these in Terminal on the spare Mac after `cd ~/josephs-board`. The `jb` script itself wasn't in the upload, so the first six come from your notes and the rest from the algo's own instructions.

Command	What it does	When to use it
`./jb status`	Lists all 7 parts and whether each is running, plus the board addresses.	First thing whenever something looks off.
`./jb restart`	Stops and starts everything. Safe to run more than once; nothing is lost.	After installing new files, or when a part shows red.

Command	What it does	When to use it
`./jb restart <name>`	Restarts just one part, e.g. `./jb restart ibkr`.	When only one tab has a problem.
`./jb logs <name>`	Shows that part's live log. Names: ibkr, webull, scanner, gateway, webull-algo, ibkr-algo. Press **Control+C** to stop watching.	When asked for "the last few lines" of an error.
`./jb webull-login`	Checks Webull API access and, if needed, approves it.	If the Webull tab can't connect or Webull asks you to re-approve.
`./jb url`	Prints the board addresses (Mac, phone, work laptop).	When you forget a link.
`./jb backtest ibkr <strategy file>`	Tests a strategy on past prices. Never touches your account. Add `--duration "30 D"` for more history or `--sessions all` to include pre-market, after-hours and overnight.	Before paper or live trading a new idea.
`./jb backtest webull <strategy file>`	Same, for the Webull algo *(from notes)*.	Same.
`./jb live ibkr <strategy file>`	Starts **real-money** trading. Asks you to type `I_UNDERSTAND_THE_RISK`. Runs in that Terminal window; Control+C stops it and cancels any open order.	Only after backtests and paper trading look good.
`./jb live webull <strategy file>`	Same, for Webull *(from notes)*.	Same.
`./jb live-reset ibkr <strategy file>`	Clears the algo's own record of its position after you've fixed a mismatch with your real IBKR position.	Only when live trading refuses to start because the positions don't match.

The 7 services

Every part starts by itself when the Mac restarts and restarts itself if it crashes. Only the gateway (8090) can be reached from other devices; the rest listen on the Mac alone and the gateway passes their data through.

Name (for `./jb logs`)	Port	What it does	Board tab
gateway	8090	The front door: serves the board, asks for the password (once per device, remembered for a year), and passes live prices from the other parts to your browser.	All of them
ibkr	8765	Streams IBKR prices from IB Gateway: stocks, ETFs and futures. Falls back to free delayed prices for symbols you don't pay for. Records every price around the clock.	IBKR
webull	8766	Streams Webull prices for stocks and ETFs (no futures). Records every price around the clock.	Webull
scanner	8767	The mover scanner: watches *your ETF list or 30 big stocks for unusual moves*.	Scanner
webull-algo	none	The Webull trading robot (paper or live) *(from notes)*.	Webull Algo
ibkr-algo	none	The IBKR trading robot. Currently in **paper** mode.	IBKR Algo
keepawake	none	Stops the Mac from going to sleep so everything keeps running *(from notes)*.	none

After a Mac reboot you must log in to **IB Gateway** yourself; the IBKR parts wait for it. Tailscale starts on its own.

Addresses

All three open the same board and ask for the same password.

Device	Address	Needs
The spare Mac itself	http://localhost:8090/	Nothing
Your phone (and any device with Tailscale)	http://100.93.57.24:8090/	Tailscale app turned on
Work laptop (no Tailscale)	https://alexs-macbook-pro.tail60bf05.ts.net/	Tailscale Funnel on. Note the **s** in alexs. This link is on the public internet, so keep it private and the password strong.

Each board can also be opened alone by adding its page name, e.g. `.../ibkr.html` or `.../webull.html`. You can also "Add to Home Screen" on the phone for an app-like icon.

Funnel commands, typed on the Mac:

- Turn on: `/Applications/Tailscale.app/Contents/MacOS/Tailscale funnel --bg 8090` (stays on after reboots)
- Check: `/Applications/Tailscale.app/Contents/MacOS/Tailscale funnel status`
- Turn off: `/Applications/Tailscale.app/Contents/MacOS/Tailscale funnel --https=443 off`

Board tabs

The top bar has five tabs. The board remembers the last tab you opened, and a tab disappears if its part is switched off in settings.env.

Tab	What you see	What you can do
IBKR	Live prices from Interactive Brokers in two groups, US EQUITIES and FUTURES. Futures follow the front-month contract and roll over automatically.	Add a symbol (e.g. `AAPL` or `ES=F`), remove one with **×**, switch **List / Tiles**.
Webull	Live prices from Webull for stocks and ETFs. Futures tiles here say to use the IBKR board. Webull streams at most 100 symbols.	Same as IBKR.

Tab	What you see	What you can do
Scanner	Mover Scanner with ETFs | Stocks buttons: your ETF list, or 30 heavily traded stocks (list in mover-scanner/stocklist.yaml). ETFs flag at a 0.5% move in 5 minutes (1% in 15); stocks at 1% (2% in 15). Shows a sortable table (Symbol, Last, Window %) and a "Recent flags" list.	Switch ETFs / Stocks (back in about 10 seconds), click a column heading to sort.
Webull Algo	The Webull robot's status, last backtest and Trading options.	Change and save Trading options.
IBKR Algo	The IBKR robot: mode (PAPER or LIVE), strategy, last price, session now, position, unrealized P&L, return since start, last order, last backtest and Trading options.	Change and save Trading options.

On every price board:

- **Ticker tape** across the top scrolls every symbol with price and change.
- **Clock** shows the time and date; under it, **equities** and **futures** status lights say when the last update came in.
- **Change boxes**: 1M, 5M, 15M, 45M, 4H, 8H, 1D, 3D, 1W, 1MO, each the % change over that window. On a phone (List view), the headline number is the 1D change; tap a row to see all boxes.
- **Bottom line** of each tile: where the price came from (`ibkr · live`, `ibkr · delayed ~15m`, `webull · live`, `webull · overnight`) and how old it is.
- **Your symbol list is shared** across all your devices, and the Mac keeps recording prices around the clock, so overnight moves are there when you look.
- **Connection** button: leave it blank. It's only for unusual direct connections.

Colour, tag and box legend

You see	Where	It means
Green dot	Tab name	That part is running.

You see	Where	It means
Red dot	Tab name	That part is not running on the Mac. Run `./jb status`.
Grey dot	Algo tab name	The robot isn't running, but old backtest results are still shown.
Green / red price number	Tile	Price is up / down over the last **1 minute**.
Green / red tint and left edge	Tile background	Price is up / down over the last **2 minutes**.
Quick green or red flash	Tile	The price just changed.
Green / red change box	1M to 1MO boxes	That window's change is up / down.
"…" in a change box	Change boxes	Not enough history yet to cover that window. It fills in by itself.
Green status light, "last update Xs ago"	Under the clock	Prices are flowing.
Red status light	Under the clock	No update for 45+ seconds, or the connection dropped.
Yellow **DELAYED** tag, price slightly faded	Next to a symbol (IBKR)	You don't pay for live data on this symbol (e.g. CL=F, GC=F), so it shows IBKR's free prices, about 10 to 20 minutes old. The algo never trades on these.
DLY	Ticker tape	Same as DELAYED.
Yellow box: "No overnight prices here…"	Top of Webull board	Your Webull plan doesn't include the overnight session (8 pm to 4 am). Regular and extended hours still work.
"no Webull futures data subscription -- see this symbol on the IBKR board"	Futures tile on Webull board	Normal. Futures come from IBKR.

You see	Where	It means
Red box at the top of a board	Price board	A real problem, e.g. "The IBKR bridge is not running on the spare Mac". It tells you which command to run.
Red text on a tile's bottom line	Tile	That one symbol has a problem (e.g. unknown symbol).
Yellow note about 100 symbols	Webull board	You've added more than Webull will stream on one App Key.
Amber row with **FLAG**	Scanner table	That ETF made an unusual move in the last 5 minutes.
Coloured lines in "Recent flags"	Scanner	Which detector fired: blue = price jump, amber = volume surge, purple = range expansion, teal = buying/selling pressure.
PAPER / **LIVE** tag	Algo tab	Whether the robot is simulating or trading real money.
Red **LIVE TRADING PAUSED** banner	Algo tab	The robot stopped itself because an order's outcome couldn't be confirmed. Check the broker app before restarting it.

Trading algos

Each robot follows a **strategy file** (a short text file of rules, e.g. "buy when the 20-minute average crosses above the 50-minute average"). The IBKR and Webull robots use the same file format, so a strategy works with either. The IBKR robot is in **paper** mode today.

The three modes, safest first

1. **Backtest**: replays past prices through the rules. Pure arithmetic; your account is never touched. Results appear on the algo tab.
2. **Paper**: runs on live prices but every order is simulated on the Mac. Nothing is sent to the broker.
3. **Live**: real orders, real money. Needs two separate switches (below).

Example strategies included (IBKR, in `ibkr-algo/strategies/`): `sma_crossover`, `rsi_reversal`, `bollinger_breakout`. Strategies trade plain US stocks and ETFs only, never futures or options.

Trading options panel (on each algo tab; changes take effect within about 5 seconds)

Setting	Default	What it does
Sessions	Regular hours only	Regular hours (9:30 am to 4:00 pm ET) are always on. Tick pre-market (4:00 to 9:30 am), after-hours (4:00 to 8:00 pm) and overnight (8:00 pm to 3:50 am, Sun to Thu nights) separately for **Paper & backtest** and for **Live**.
Order type	Limit	Limit or Market. Market orders go out only 9:30 am to 4:00 pm; outside those hours a limit order is sent instead.
Limit price	0.05%	How far from the last price to set the limit so it fills quickly. A negative number waits for a better price (entries only).
Cancel if not filled after	60 seconds	An unfilled order is cancelled and shown as "unfilled", not an error. The robot tries again if the rule still holds.
Trailing stop	Off	Exits when the price falls this % from its best since entry. Watched by the robot, not held at the broker, so it only works while the robot runs.
Extra cost outside regular hours	0.1% per fill	Paper and backtests only, to account for wider spreads.
Use the strategy file's settings		Clears the panel so the strategy file (or built-in defaults) applies again.

Which setting wins: the panel first, then the strategy file's `trading:` section, then built-in defaults.

Safety rules (live trading)

- Two switches must both be on: `confirm_live: true` plus a maximum order value in the strategy file, **and** typing `I_UNDERSTAND_THE_RISK` when you run `./jb live`.
- Every order is checked against a share limit and the maximum order value.
- It refuses to start if your real position differs from its own record, or an order is still open.
- One order at a time; only what the broker reports as filled is counted.
- A rejected order means no new order for 15 minutes; 3 rejections in a row stop it.
- If it can't confirm what happened to an order, it pauses and shows a red banner.
- It never trades on delayed (yellow DELAYED) prices, in paper or live mode, and says so on the IBKR Algo tab.
- Control+C in the live window stops it and cancels any open order.
- Start small, watch the first trades, and always check fills in the broker's own app.

settings.env switches

`~/josephs-board/settings.env` holds your private keys and on/off switches. Only the names are listed here. Never paste this file into a chat, and run `./jb restart` after changing it.

Name	What it controls
`JB_PASSWORD`	The board password (secret).
`WEBULL_APP_KEY`, `WEBULL_APP_SECRET`	Your Webull API keys for the real account (secret).
`RUN_WEBULL_BRIDGE`	Turns the Webull price feed on or off *(from notes)*.
`RUN_WEBULL_ALGO`	Turns the Webull robot on or off *(from notes)*.
`WEBULL_ALGO_STRATEGY`	Which strategy file the Webull robot runs *(from notes)*.
`RUN_IBKR_ALGO`	Turns the IBKR robot (paper) on or off.
`IBKR_ALGO_STRATEGY`	Which strategy file the IBKR robot runs, e.g. `strategies/sma_crossover.yaml`.

Name	What it controls
`IBKR_ACCOUNT_ID`	Only needed if one IBKR login has several accounts.
`IBKR_GATEWAY_PORT`	The port IB Gateway listens on. Leave as set up.
`WEBULL_ENVIRONMENT`, `WEBULL_REGION_ID`	Which Webull system and region to use. Leave as set up.

Not a setting to put in the file: `IBKR_LIVE_TRADING_CONFIRM` is set only for one Terminal window when you type the confirmation in `./jb live`. Other names you may see (bridge tokens, data folder, bind addresses) are internal plumbing; leave them alone.

Folders, logs and backups

Everything lives in `~/josephs-board` on the spare Mac.

Folder or file	What's in it
`jb`	The control command.
`settings.env`	Keys, password and switches (private).
`install.sh`, `requirements.txt`	The installer and its list of parts. Already run.
`gateway/`	The front door (port 8090).
`bridges/`	The IBKR and Webull price feeds.
`mover-scanner/`	The Scanner.
`webull-algo/`, `ibkr-algo/`	The two robots, their `strategies/` folders, `logs/`, and `trading_options.json` once you save the panel. Backtest trade logs and charts land next to the strategy file.
`web/`	The board pages your browser shows.
`lib/`, `bin/`	Shared helpers and service starters.
`data/`	Your shared symbol list and the price history the Mac records (about 35 days).
`logs/`	Each service's log, read with `./jb logs <name>`.

Folder or file	What's in it
`backup-delayed/`	Your files from before the delayed-prices fix (Oct 5).
`backup-webull/`	Your files from before the Webull overnight/futures fix (Oct 5).

To undo the delayed-prices fix: `cd ~/josephs-board && cp backup-delayed/ibkr_bridge.py bridges/ && cp backup-delayed/board.js backup-delayed/board.css web/ && cp backup-delayed/paper_trader.py ibkr-algo/algo/ && ./jb restart`

In this project's files: `JOSEPHS_BOARD_v3.zip` (the original full code), `Josephs_Board_Setup_Guide.pdf` (setup Steps 1 to 11), and `josephs-board/` (the Oct 5 fixes and their steps).

Overnight features (October 2026 updates)

Everything below was added in five updates. Each page also explains it under **? Legend**.

Feature	Where	What it does
VS 4PM CLOSE · OVERNIGHT · 9:30 GAP boxes	IBKR and Webull boards	Change since the 4 pm close, the 8 pm – 4 am overnight move, and today's opening gap. The 1D box is now called **24H** (the last 24 hours).
O/N range bar, 1W and 52W bars	Boards	Overnight low to high with a dot at the price now. Green dot = above the overnight high, red = below the low. Under it, two smaller bars: 1W (last 7 days) and 52W (last year), low to high with a dot at the price now, counting after-hours and overnight too. Green dot = at a new high for that range, red = at a new low. Hover or long-press for the dates.

Feature	Where	What it does
Session badge	Under the clock	Which session it is in New York and how long until the next one (holidays aren't known).
Ticker tape futures	Top of every board	ES, NQ, YM grouped after a yellow FUTURES label with their change vs. close (Webull gets them from IBKR).
A–Z order, faded Webull tiles	Boards	Symbols listed alphabetically. Webull stock tiles fade 8 pm – 4 am because the Webull plan has no overnight prices.
US EQUITIES ▾ menu	Boards	Show only holding, with alerts, or moving overnight ±1%. Nothing is removed.
Alerts	Boards, Alerts button	Price above/below or % move vs. close, checked by the Mac day and night, pushed to the phone through the free ntfy app.
Morning cards	Top of the boards, phone	7:00 and 9:00 am ET weekdays: futures, biggest moves, overnight high/low breaks. Copy for Claude button.
Recap video	Recap tab; sent to phone (ntfy) at 8:05 am ET on trading days	2-3 minute narrated video since the 4 pm close: after-hours (SMART), overnight (Blue Ocean), pre-market (SMART), then where things stand with your holdings. Make one now button; time set by JB_RECAP_TIME in settings.env (or off). Last 14 kept in data/recaps/.
Chart lines and markers	Price chart pop-out	4 pm close (yellow), overnight high/low (purple), news ▼, Compare line, volume bars, your cost (pink), your buys and sells.
Since 4 pm close, bold keywords	News tab	Only news after the last close; words like earnings or FDA in bold yellow.

Feature	Where	What it does
Vs close column, session chips, volume ratios	Scanner, Movers	Change vs. close, flags filtered by session, and volume vs. this session and vs. the last hour.
Overnight picks, earnings, track record	Scanner, Picks	Overnight setups, next earnings dates (Nasdaq), news in the last hour, and a daily scored pick history.
Portfolio tab	Portfolio	Type in each buy and sell; positions with average cost and gain/loss; HELD line on the boards; basket backtest vs. SPY (Yahoo daily prices).
Overnight gap strategy	IBKR Algo (paper)	Buys 7:30 – 8 pm ET Mon–Thu if up 0.5%+ vs. close, sells at the 9:30 open; 2% stop, 4% take-profit. File: ibkr-algo/strategies/overnight_gap.yaml.
Run backtest button	Both algo tabs	Pick a strategy and run a backtest on the last 10 days of 1-minute prices, no Terminal.
Paper trading report	Both algo tabs	One line per finished trade with sessions, entry, exit, P&L and running total, grouped by day, with a line chart.
Stop & close position	Both algo tabs	Kill switch: closes the position and stops new trades until Resume trading. Holds through restarts.
Webull auto-reconnect	Webull bridge	Reconnects by itself after the 417 INVALID_SESSION error.

New files and data: gateway/watch.py, portfolio.py, algoctl.py; data/alerts.json, morning.json, holdings.json, ibkr_volume.json, webull_volume.json, picks_history.json, earnings.json; JB_NTFY_TOPIC in settings.env. Backups of each update are in folders named backup-overnight, backup-watch, backup-scanner2, backup-portfolio and backup-algo.

World, Signals, Analysis, My trading and Extras

Five tabs and extras added on Oct 6, all free sources. Each tab's **? Legend** explains it on the page.

Feature	Where	What it does
World tab	After Webull	Overnight mood meter (-100 to +100), market clock, futures, VIX, rates, dollar, Asia and Europe, crypto, economic calendar (Fed, data, Treasury auctions). Sunday evening push when futures open. Prices from Yahoo.
Signals tab	After News	Phone-alert switches: big or fast move, heavy volume, yesterday high/low, 4 am gap list, StockTwits buzz, SEC 8-K / insider buy / 13D, earnings move, 7:30 am overnight summary.
WHY? button, EARNINGS / BUZZ / 8-K tags	Board tiles	WHY? lists the likely reasons a stock is moving (news, filings, earnings).
Analysis tab	After Signals	Gap scorecard (how often gaps fill), session split (after-hours / overnight / pre-market / regular), sector heatmap, relative strength vs SPY and QQQ, stocks that move together, options max pain and put/call ratio.
Chart indicators	Price chart pop-out	VWAP, MA 20, MA 50, RSI, previous day high/low buttons.
IBKR 366 note	Under board tiles	The "No historical data query found (366)" message now says why it is harmless.

Feature	Where	What it does
My trading tab	After Analysis	Journal (IBKR fills every 5 min, Webull every 10, or by hand) with tags and notes; win/loss by session, tag and stock; daily totals; size calculator; daily loss limit push with optional algo pause; paper algo leaderboard (8:30 pm ET); replay a recorded night.
Weekly recap video	Recap tab, phone	Fridays 8:15 pm ET (JB_WEEKLY_TIME), or Make weekly one now: week's best and worst, market weather, Picks and your trades, best paper strategy, next week's earnings and big events. Last 8 kept.
Market weather	Recaps, phone page	Each sector fund gets sunny / partly sunny / cloudy / rainy / stormy from its move since the close (2.5x wider steps for the week).
Health tab	Last tab	What is running, when each broker last sent a price, disk space, last log lines, and a Restart button per part (same as ./jb restart).
Phone page	Phone link, top bar	One screen: mood, futures, weather, biggest moves, holdings, today's trading, alerts, coming up. Safari: Share, Add to Home Screen.

Feature	Where	What it does
Look button: Polished / Pure black / Clean light / Night / Auto	Button, top right	Tap it (top right of the board and the phone page) for a list. Polished is the normal look: calmer green and red, rounded tiles with a little space between them on a coloured area, AH badge, clean font for labels and buttons. Pure black and Clean light (grey and white) are polished too. Night = dim red, on every page, and the phone's bottom bar stays. (Classic and Old light were taken out on Oct 9: they copied Polished and Clean light.) Auto = Night 9 pm to 6 am, Polished otherwise. Only colours change, never what the board does. Remembered per phone or computer. If tiles look cramped on the Mac, press Command+0 (Safari zoomed out).

New files and data: gateway/world.py, signals.py, analysis.py, trading.py, extras.py; web/world.html, signals.html, analysis.html, trading.html, health.html, phone.html, theme.js; data/world.json, signals_log.json, analysis_cache.json, gap_log.json, journal.json, leaderboard.json, weather.json. Backups: backup-world, backup-signals, backup-analysis, backup-trading, backup-trading-fix, backup-extras.

If a part shows red, open the **Health** tab and press Restart before using Terminal.

SELL ALL, phone fixes and futures pop-up

Added on Oct 6. SELL ALL is an emergency button that sells everything on one broker; it passed a live test on both IBKR and Webull (1 AGNC share each). Nothing else on the board was removed.

Feature	Where	What it does
SELL ALL button	IBKR and Webull boards, next to Connection; Emergency card on the Phone page	Opens a window showing the account (paper or LIVE), every position and open order, and how it will sell now. Nothing is sent until you type SELL ALL and press the red button.
What SELL ALL does	That broker only	Pauses that broker's algo, cancels every open order, then closes every stock (and IBKR futures). Options are left alone. You get a phone push when it finishes.
How it sells	By session	Regular hours: market orders. Pre-market, after-hours, overnight: limit orders 1% through the last price, then 2%, then 3%, 20 seconds each. Markets closed: cancels and pauses only.
After SELL ALL	Algo tabs	The algo stays paused until you press Resume on its tab.
Webull account	settings.env	WEBULL_ACCOUNT_ID picks the Individual Margin account (your keys also see an Events Cash account).
Webull speed limit	Automatic	If Webull says "too many requests", SELL ALL waits a few seconds and asks again (jb-killfix).
Tabs list	Phone, top-left of the tab bar	The ☰ Tabs button lists every tab, so you don't have to scroll sideways.
Phone fixes	Every page on a phone	Typing no longer zooms the iPhone in; buttons and checkboxes are bigger; board header text is larger; wide tables keep the symbol column in place, with a "swipe sideways for more" hint. Computer screens are unchanged.

Feature	Where	What it does
Futures pop-up	Phone page, Futures and fear tiles	Tap a tile to see the price, change vs the 4 pm close, since yesterday, the 24-hour high and low, and a 24-hour chart.

New files on the Mac: gateway/killswitch.py, web/kill.js, web/mobile.js, web/phonefut.js. Backups are in backup-kill/, backup-phonefut/ and backup-killfix/.

Final ideas (Oct 7): orders, ETF edge, feeds, guardian, tools

Ideas 1-72 from the final-ideas list (all except 21), installed with jb-ideas. Most parts are new sections at the bottom of the **Night desk**; nothing else on the page moved. Night desk guide part 2 (pages 9 onward of the guide PDF) has the details.

Part	Where	What it does
Safety stops, targets, trailing, break-even	Night desk; ticket **More**	The board watches the price in every session and sells (stepping 0.5% to 3% under the price) when hit. Watched by the board, not placed at IBKR
Ticket extras	Order ticket	Good for: session / overnight + tomorrow / until cancelled; ladder; dollar size buttons; IBKR preview; duplicate warning; Flip; Chase
Scheduled orders	Night desk	At the open, 4 am, 4 pm, 8 pm or a set time, with optional conditions; paper by default; sell all at the open
Fill pushes and notes	Phone; My trading	A push per fill with the distance from your limit or last price; an automatic note per fill; scorecard; weekly PDF report Friday 8:30 pm

Part	Where	What it does
Second chip row	IBKR and Webull boards	Trend light, FAIR, BUYERS / SELLERS, PRICES DISAGREE, REDDIT #, NOTE, the fund card
Options lines	QQQ / SPY charts	Options magnet, fast and gamma flip levels as dotted lines
World additions	World tab	VIX curve warning, Fed meeting odds, Nasdaq-100 breadth, Bitcoin order book (Gemini)
3x ETF edge	Night desk	Fair value, rebalance, decay, drift, opex calendar, net liquidity, Treasury auctions, CFTC, yen and VIX pushes, price history
Feeds	Night desk	Top news ranked, IBKR newswire, Reddit, Bluesky, Polymarket, policy posts, buy / sell pressure, IBKR vs Webull cross-check
Mac guardian	Night desk; System check; Health	Power-cut restart, memory, clock, macOS updates, crash loops, backups 2:30 am, internet drops, SELL ALL re-check, Sunday self-test, Saturday refresh
Algo safety	Both algos	Heartbeat, skip entries when prices disagree; live only: position check, safe mode, order speed limit
Command bar and keys	Top bar ⌕ or /; Phone search box	TQQQ, TQQQ buy, TQQQ news, flip SOXL, tab names; keys 1-9, 0, B, S, Esc, ?
Any-symbol chart	Command bar	Any symbol, future or index in the board's own chart (Yahoo prices)

Part	Where	What it does
Filled-in chart	Charts	Stretches filled from IBKR after downtime, drawn in a dimmer blue-grey
Notes, what changed, widget, Siri, alert manager, go-live	Night desk; board and Phone	Symbol notes, a since-you-last-looked box, iPhone widget and Siri shortcut, every alert on one page, go-live checklist (./jb live runs it)

New files in ~/josephs-board: gateway/orders.py, edge.py, feeds.py, guardian.py, tools.py, iblink.py; bin/golive.py; web/nd-orders.js, nd-edge.js, nd-feeds.js, nd-guardian.js, nd-tools.js, plus2.js, world-plus.js, tools-ui.js, trading-plus.js, health-plus.js. New folders: backups/ and web/reports/. Backup of the old files: backup-jb-ideas/.

Sentiment report and phone link boxes

Added on Oct 7 (jb-sentiment). Both are explained in the board's **? Legend**.

Where	What it does
MACRO tab: Market sentiment report	**Generate report** makes one report on the market's mood by fixed rules from free data the board already has. Short term (next 1-3 days): overnight mood meter, VIX level and move, VIX curve, last 5 days, today's breadth, 2 days of macro headlines. Mid term (next 2-8 weeks): S&P and Nasdaq vs their 50-day average, last 20 days, Nasdaq-100 stocks above their 50-day, Fed odds, hedge fund positioning, a week of headlines, net liquidity. Long term (next 3-12 months): the MACRO score, S&P vs its 200-day average, 50-day vs 200-day, 52-week range, last 12 months, 6 months of net liquidity.

Where	What it does
Scores	Each line scores -1 (bearish) to +1 (bullish); a time frame is their average x 100; the combined score is the three averaged equally. Bullish +40 or more, Leaning bullish +15, Neutral, Leaning bearish -15, Bearish -40 or less. Also: one sentence on how the time frames fit together, a line for 3x bull vs 3x inverse funds, and big events in the next 3 days. Rules, not advice.
Report buttons	Copy text; Print / Save as PDF (prints just the report); Ask Claude about this report; Earlier reports lists the last 10 (60 kept on the Mac).
Phone page: link boxes	All ten bottom links as equal cyan boxes, two per row: Night Desk, Order Ticket, IBKR Algo, Webull Algo, Full Board, Macro, Ask Claude, Recap Video, System Check, Health.

New files: gateway/sentiment.py, web/sentiment.js; web/phone-links.js replaced. Backup in backup-jb-sentiment/.

If something looks wrong

Try these in order. If one shows an error, send only its last few lines.

What you see	Try this
A red dot or red box on a board	`./jb status`, then `./jb restart`.
IBKR tab red after the Mac restarted	Log in to IB Gateway on the Mac, then wait a minute.
Webull tab can't connect	`./jb webull-login`.
Board asks for a password again	Normal on a new device or after clearing the browser. Enter the board password.
Phone can't open the board	Check the Tailscale app is on, on both the phone and the Mac.

What you see	Try this
Work laptop can't open the link	Check the **s** in alexs, then run the Funnel status command (Addresses section). If it's off, run the turn-on command.
A tile shows DELAYED	Normal for symbols you don't pay for. Buy the IBKR data and run `./jb restart` to make it live.
Change boxes show "…"	Wait; they fill in as history builds.
Algo tab shows LIVE TRADING PAUSED	Check your position and orders in the broker's app before doing anything else.
Anything else	`./jb logs <name>` for the part with the problem, and send the last few lines.

Phone bar, Chart and Report Card (jb-reportcard)

Three additions from October 9, 2026: easier page moves on the phone, a chart for any ticker, and a Report Card where the board grades its own calls.

Feature	Where	What it does
Phone bar	Bottom of every page on a phone	**Back** (the page or tab you came from; on the full board it steps back through the tabs you opened), **Home** (the Phone page), **Chart**, **Menu** (every page and tab as big boxes). Computers are unchanged.

Feature	Where	What it does
Chart tab	Full board tab, and Chart in the phone bar	Type any ticker (TQQQ, AAPL, NQ=F, ^N225) for its line chart over 1D, 5D, 1M, 6M, 1Y, 5Y or All (its full history, one point a month). Prices from IBKR first, including the overnight venue; Yahoo when IBKR can't (indexes, crypto, stocks your IBKR plan doesn't cover). Shaded with a paler line = outside regular hours; cyan dashed = 9:30 am open; grey dashed = price at the start of the range. Your last 8 symbols are buttons.
Chart order flow	Chart tab at 1D, the Order flow buttons	Four extras, each with its own on/off button (turn all off for a clean chart). Buy/Sell line: shares bought at the ask minus shares sold at the bid, added up since 8 pm ET (dashed part = estimated from 1-minute bars before the page started watching). Big trades: 10x the day's average size and worth $100k+, circled on the line and listed. Volume at price: side bars of shares traded at each price today, busiest in amber. Bid/Ask: the best bid and ask now with shares on each side. Free IBKR data, so only symbols with live IBKR prices, up to 3 at a time; works pre-market, regular, after-hours and overnight. Not the full order book (that needs paid Level 2).

Feature	Where	What it does
Report Card tab	Full board tab, and Report Card in the phone menu	How often each feature was right: mood meters, Picks and each of their signals, ETF pick, gap scorecard, 4 am gaps, alerts, sentiment report, MACRO score, earnings expected moves, max pain, put/call, politicians' trades, relative strength, market weather and event days. Green 55%+ right, amber 45-55%, red under 45%, grey = fewer than 20 graded. 50% = a coin flip.
Live vs Replay	Each Report Card card	Live = calls saved since install, graded with free Yahoo prices 30 and 60 minutes later, at the next open and close, and 1, 5, 20 and 60 days later. Replay = the same rules re-run on up to 2 years of past prices, for price-only features; it runs each evening and takes a few minutes.
Card details	Tap a card	Each part (meter window or signal), each session the call was made in, a line of accuracy over time, and the latest calls with a check or cross.
Your trades vs the board	Bottom of the Report Card	Win rate and average result of your journal trades when the Futures mood agreed with you, was against you, or had no clear call. Inverse ETFs count the other way round.
right 61% tags	Next to the Futures mood, Picks mood and Mood since the 4 pm close meters	How often that meter, over the window you picked, pointed the right way by the next close (the open for the 4 pm mood), with how many readings. Tap to open the Report Card.

Feature	Where	What it does
Weekly push	Phone, Fridays 8:30 pm New York	The week's best and worst feature. Switch it off at the bottom of the Report Card.

Roku TV view

Added on Oct 9: a read-only TV app called **Alex Hale's Board** for the Roku, sideloaded with Roku's free developer mode. The board's **? Legend** explains it too.

Where	What it does
Top row	Market mood, Futures mood and Picks mood meters (since the 9:30 open during regular hours, since the 4 pm close otherwise).
Middle: IBKR tiles	The IBKR board's stocks, then its futures, in two labelled groups, as tiles like the IBKR board: they flash green or red as the price moves, the tint and side bar follow the last 2 minutes, boxes show the same moves as the IBKR board (1M, 5M, 15M, 45M, 4H, 8H on one row; 1D, 3D, 1W, 1MO, O/N on the next), and the O/N, 1W and 52W range sliders sit under each. Prices every 2 seconds, up to 24 tiles, in the IBKR board's own order.
Bottom	Top movers, a Futures 24 h chart (one coloured line per IBKR future showing its % change over the last 24 hours, with a key listing them best first), Picks and Coming up (big reports plus the latest alerts) as lines of text, then the market weather across the full width. The rest refreshes every 15 seconds.
Read only	The TV can only look: no trading, SELL ALL, algo pause or settings, and no account money or positions on screen. It uses its own TV key (data/tv_key.txt), not the board password, and that key opens only the TV summary.

Where	What it does
Where it works	Home Wi-Fi first, then the ts.net link (any Wi-Fi). The top line says which one it is using.
Open it	Press Home on the Roku remote and pick Alex Hale's Board (at the end of the channels; move it with the star button, Move channel).
OK button	Reconnect now. It only asks the board again and changes nothing. If the Mac's Wi-Fi address changes, the TV learns the new one by itself through ts.net.
Last resort	If it still can't reach the Mac: check the Mac is on and awake, then run `bash ~/josephs-board/roku/make-tv-app.sh` and send the new Board-TV.zip from Downloads to the Roku (Roku at 192.168.1.145, user rokudev).
Lock out the TV	Delete data/tv_key.txt, restart the gateway, and make the TV app again.

New files: gateway/tvview.py (the TV feed, /api/tv), roku/app (the Roku app), roku/make-tv-app.sh (makes Board-TV.zip). Backup: backup-jb-roku/.

Alex's channel

Added on Oct 9: your board as a TV news channel, with an anchor's voice reading what is on screen. It plays on the Roku, the phone and the Mac. The board's **? Legend** explains it too.

Where	What it does
The show	Segments in turn: BREAKING (when a board alert fires), the opening with your three mood meters, futures with a chart that draws itself, Top movers, Picks with their reasons, news for your symbols, sectors, Coming up and the market weather. A ticker crawls along the bottom, and the corner shows ES, NQ and the clock.

Where	What it does
The anchor	The Mac's own free voice (the Recap voice, Samantha unless changed) reads each segment while it is on screen, so the voice always matches the picture. When the sound is off, the same words show as closed captions near the bottom of the screen, two lines at a time. A fresh show is made about every 2 1/2 minutes while someone is watching, and none at all when nobody is.
On the TV	In the Roku app, the right arrow opens the channel and the left arrow (or Back) returns to the board. In the channel, OK or Play turns the sound off and on, and the right arrow skips to the next segment.
On the phone and Mac	Click the small ▶ Channel button next to the look button (top right of the full board and the Phone page), open channel.html on the board, tap Watch Alex's Channel on the Phone page, or pick Alex's Channel in the phone Menu. Tap Tune in with sound, since phones only play sound after a tap. Turn the phone sideways for a bigger picture.
Read only	It only reads what the board already shows. No account money or positions are shown or spoken. The TV uses its TV key and the phone and Mac use the board login.
Voice settings	JB_CHANNEL_VOICE and JB_CHANNEL_RATE in settings.env give the channel its own voice. Otherwise it uses JB_RECAP_VOICE and JB_RECAP_RATE. Sound files are kept in data/channel/ and old ones are deleted by themselves.

New files: gateway/channel.py, web/channel.html, web/channel-link.js, and the Roku app's ChannelView.brs and ChannelTask files.

A deeper channel with natural voices (batch I)

Added with batch I (jb-anchor): the channel became a longer, deeper show with two natural-sounding anchors. The board's **? Legend** explains it too.

Where	What it does
Deep dives	After the headlines come 2 or 3 deep dives, taken in turn from a pool so the next loop is different. A full cycle runs about 10 to 15 minutes before anything repeats. The pool: one of your 3x funds in depth, futures in depth, sector rotation, rates, dollar, yen and oil, the calendar with earnings, the desk view, Asia and Europe, big traders' positions (CFTC), crypto, and Fed odds. Each one has a 24 hour chart with level lines when there is one, chips and short points.
Why, normal, next	Every segment says why (linked causes, plus the board's own move breakdown from batch C when it is installed), compared to normal (vs its usual overnight move, days in a row, where it sits in its 1-week and 52-week range), the session story (after-hours, overnight, pre-market) and what to watch (overnight high and low, the 4 pm close, the options magnet, the next big event).
What changed	The anchor remembers what it last said. If a number barely moved it says "still" or skips it. If it moved it leads with that, for example "Since my last update 9 minutes ago, Nasdaq futures slipped another half percent". Numbers are rounded the way people say them, companies are said by name (Nvidia, not N V D A), there are short pauses between stories and a little stress on key words, and each line has several wordings.
Show schedule	The show name sits next to LIVE. Headline Hour at the top of each hour, Quick Update at :15, :30 and :45 (only what changed), Closing Bell Wrap at 4 pm, Overnight Open at 8 pm, Asia and Europe Check at midnight, QUIET HOURS from 1 to 3 am, Pre-market Preview at 4 am, Opening Bell Preview at 9:15, Sunday Futures Open at 6 pm Sunday, and Weekend Desk when markets are closed.

Where	What it does
QUIET HOURS	From 1 to 3 am ET a big purple QUIET HOURS band with a moon stays at the top of the screen for the whole show, and the anchor says it is quiet hours. The deep dives are swapped for short explainers built from your board's own numbers: why 3x funds lose value in choppy weeks, how often overnight gaps hold into the open, how much of the move happens overnight vs in the day, and how often there are big days. The Weekend Desk shows the same band with one explainer.
Specials	Up to 3 hours before a big report (CPI, jobs, the Fed, PCE, GDP, retail sales) a SPECIAL REPORT shows a countdown, the forecast and last time. In the hour after it, WHAT JUST HAPPENED shows how Nasdaq and S&P futures moved since. The same happens for earnings of your own symbols.
Desk view	THE DESK VIEW gives the board's rules in one line (leaning risk-on, risk-off or neutral, and whether the meters agree), with its Report Card record ("right 61 percent of the time") and whether its last graded call was right. Rules only, not advice.
Two anchors	Anchor 1 reads markets, futures, movers, picks and deep dives. Anchor 2 reads the news, coming up, specials and explainers. Natural voices are free Piper voices (Lessac, Ryan and Amy) made on the Mac itself with no internet. If they are not installed, too slow on this Mac, or ever fail, the channel uses the Mac voice (Samantha).
Voices page	channel-voices.html, or Voices at the top of the channel: try each voice with ▶ Sample, then pick both anchors, the speed and the chime, and press Save.
Chimes	A short chime made on the Mac starts each story, a longer one for BREAKING and a softer one for explainers. Nothing to license.

New files: web/channel-voices.html, bin/jb-piper.py, bin/jb-voices-setup.sh (run it again to re-download the voices), and the voices/ folder. Changed: gateway/channel.py, web/channel.html and the Roku app's ChannelView.brs. Backup: backup-jb-anchor/.

Deep search upgrades (batches A to H)

Added on Oct 9, 2026 from the deep search list, installed in order A to H. Everything is free, and nothing here sends a real order by itself. The board's **? Legend** explains each one on the page.

Where	What it does
Accounts and protection (A)	Phone page card and Health tab: for IBKR and Webull, the account type (both are margin accounts), buying power and margin call risk. IBKR cushion under 25% is amber, under 10% is red with a push. Each position shows AT BROKER (a real stop at the broker), MAC ONLY (a board stop that stops if the Mac goes down) or NO STOP. Place at IBKR sends a real stop-limit order that works 4 am to 8 pm.
Restart from the page (A)	The red not-running line on the Phone page and the red banner on the IBKR and Webull boards have a Restart button for each part.
Mac not reachable (A)	Opened from the https ts.net link, the Phone page keeps its last copy. If the Mac is asleep, has no power or no internet, you see that copy with a red Mac not reachable line and its time.
Backup internet (A)	Health tab: type your iPhone Personal Hotspot name and password. If the home internet is down for 90 seconds, the Mac joins the hotspot, pushes you, and tries home Wi-Fi again every 15 minutes.
Overnight session check (A)	On Sunday Dec 6, 2026 at 9:20 pm and 12:30 am the board checks overnight prices still flow and IBKR still uses its OVERNIGHT route, and pushes the result.

Where	What it does
Other fixes (A)	Report Card grades come in on time again, futures order flow starts at the 6 pm Sunday open, the Webull paper algo trades TQQQ, and the margin wording on FLIP is correct.
Exposure and exit plans (B)	My trading tab and a Phone card. Real exposure counts what you hold in index terms ($10,000 of TQQQ = $30,000 of Nasdaq-100). Cool-down suggests a break after 2 losses in a row or half the loss limit; it never blocks. Earnings moves for your 2x and 3x funds.
Exit plans (B)	Named plans such as Overnight scalp: sell a third at +2%, a third at +4%, trail the rest 2%, stop -2%. Pick one in the order ticket's More section or on a holding.
Wash-sale watch (B)	A note on the ticket when a buy would postpone a recent loss for taxes, plus the year's total. An estimate, information only.
Overnight price record (B)	After-hours, overnight and pre-market minutes are kept for good in data/history (about 10 MB a month), with Download CSV. Replay plays back a recorded night.
Your trades on the chart (B)	Your own fills appear as small green and red arrows on the Chart tab.
Why row (C)	A row on each tile from the board's own math: four trend arrows (5-minute, hourly, daily, weekly), the move split into market, sector and its own, volume vs the same time on past days (VOL 3x or THIN) and the last key level crossed. The same lines show on the Phone page, the WHY panel and in pushes.
Board math on the Chart (C)	S/R lines (support and resistance), simple patterns (double top, flags, wedges) and a Why it moved box under the chart. Each can be turned off.

Where	What it does
Scanner why tags (D)	Each mover gets up to 3 tags (OWN MOVE, MARKET-LED, SECTOR-LED, HEAVY VOL, THIN, BREAKOUT, STRETCHED, TRENDING), a summary line above the table, filter buttons, and six new scanner checks (breakout on volume, VWAP cross, squeeze release, new high on falling volume, 3 strong, fade).
Easier phone page (E)	Jump chips under the search box, cards that fold with a tap, bigger buttons and text, and tab colours by family on the computer.
Report Card summary (E)	The 3 features right most often and the 3 least, above the cards, with a Sort switch.
Friendlier pushes (E)	Pushes read like a person with the same numbers, for example Heads up: SOXL just dropped 3.1% in the last 15 minutes.
Lock Screen widget and menu bar (E)	The Scriptable script also fits the iPhone Lock Screen: S&P, Nasdaq and Dow futures and a health dot. A dot in the Mac menu bar (free xbar or BitBar app) shows the board's health and each data feed.
Desk notes (F)	A Phone card with a short note for what you hold and the biggest moves, written from the board's own numbers. The WHY panel shows the same note.
Crowd line (F)	CNN Fear & Greed, AAII and NAAIM in one line under the World mood meter and under Futures and fear on the Phone page, with a push at extremes.
Analysis articles (F)	Nasdaq.com analysis articles on the News tab with an ANALYSIS tag and an Analysis filter chip.

Where	What it does
Report Card explorer (G)	Explorer, backtest at the top of the Report Card: every call with filters, a likely range, splits by session, symbol and weekday, and Download CSV. Backtest and Monte Carlo treat each up or down call as a trade, per feature and for the board as a whole, with a luck test. It reruns each evening at 7:15 pm ET.
Rules (H)	Phone card and rules.html: when a price crosses a level, a symbol moves X% since the 4 pm close, a why tag shows up, the Futures mood turns, or a time comes, then send a push, open the ticket filled in (you tap Send yourself) or place a paper order tracked with its profit or loss.
Strategy lab (H)	Buttons next to Run backtest on each algo tab. Build a strategy by picking conditions from lists; it is saved as strategies/lab-name.yaml and is always paper.
Robustness test (H)	About 35 backtests on the board's own minute record, 10 to 20 minutes in the background: a grid of two settings from 60% to 140%, walk-forward (tuned on the older 70%, tested on the newer 30%) and Monte Carlo. Mostly green = robust, one green square = luck.

New files: gateway/safety.py, tradetools.py, why.py, scanwhy.py, insight.py, rc_explore.py, rules.py, stratlab.py; lib/jb_robust.py; web/safety.js, tradetools.js, why.js, movers-why.js, phone-ease.js, tabgroups.js, rc-summary.js, insight.js, explorer.html, lab.html, rules.html, rules-ui.js, algo-lab.js; menubar/jb-health.1m.py. Backups: backup-jb-safety/ to backup-jb-lab/.

Moved and merged (October 10)

Added on October 10 with jb-desk. Nothing was deleted: every page still opens on its own, and old links and bookmarks jump to the new spot. The **? Legend** explains each part on its page.

Where	What it does
Scanner: Top 5	The board's five strongest picks right now, ranked from its own readings. Each card shows the lean (Up or Down), price, move Since Picked, Vs 4 PM Close, a small line, Signals Agreeing and a Performance Rank. The board records every call and grades itself after 30 minutes, 1 hour, 4 hours and 1 day, with a letter grade once it has 10 calls. How The Board Did Over Time and Recent Calls sit below. Saved in data/top5.json for 45 days.
Scanner: Ask Claude	A stripped back Ask Claude box. Type a question and press Ask Claude With Live Data: it always adds the scanner rows, flags, picks and Top 5 as they are at that moment, then opens Claude. Copy Only copies the same text. It is a fresh snapshot each press, not a live stream (a stream would need the paid API).
Scanner: ETF Pick, Gappers, Twins	The ETF pick of the hour (with its push), the pre market gappers and the leveraged ETF twins from the Night Desk, shown again as Scanner views. The Night Desk keeps its own copies.
Scanner: search box	The For Search Only box is hidden.
Macro	Fed meeting odds for the next 4 meetings, Nasdaq breadth, the Bitcoin order book and the economic calendar moved here from World. World shows a note pointing to Macro.
Signals	Signals and Health are now one tab with an on-page switch: Signals or Health.
Portfolio	Portfolio and My Trading are now one tab with an on-page switch: Portfolio or My Trading. The Add a Trade boxes line up evenly.
Night Desk	The Recap Video player sits near the top (Make One Now, past recaps, Recap Page link), so Recap is no longer its own tab. Holding Night Vs Day now comes before Overnight Venue.

Where	What it does
Webull algo and IBKR algo	Plain-English Algo Notes: two boxes, Algorithm Themes and Backtesting Ideas, saved as you type. Copy for Claude: Build It and Copy for Claude: Check Before We Run copy your notes with the strategy, last backtest and paper report, then open Claude. Paste the strategy Claude wrote, then Check and Save (Paper Only) saves it as strategies/claude-name.yaml, always paper. Nothing else on the tab changed.
Page titles	Page names such as Macro and Night Desk use a formal engraved font (Cinzel, free license) with a soft gradient: silver on the dark looks, navy on Clean light, red at Night.
Scanner: Politics	Politicians' trades and Big policy posts (Truth Social, White House) on their own full-width Scanner view. They are the same parts as on the Night Desk, so switches changed on one page change on the other.
Portfolio: From My Accounts	Your Positions fill themselves every 5 minutes from your IBKR account and your Webull Individual Margin account (shares and average cost, read only). Each row is tagged IBKR, Webull, or Typed for a trade you entered that neither account holds. The bar above the list shows when each account was last read, with Refresh Now and Turn Off (back to only your typed-in trades). Saved in data/broker_positions.json.

New files: gateway/top5.py, gateway/algonotes.py, web/scanner-plus.js, web/algo-notes.js, web/subtabs.js, web/tabmerge.js, web/nd-only.js, web/nd-recap.js, web/titles.js, web/titles.css, web/fonts/. Changed: world-plus.js, plus.py, index.html, signals, portfolio, health, trading, macro, movers, night and algo pages, looks.js, legend.js. Backup: backup-jb-desk/.

Phone (October 10)

Added on October 10 with jb-phone. These change the phone only; the Mac board looks the same.

Where	What it does
Phone page buttons	Channel, Board and the look button are the same size and colour. Board opens the full desktop board on the phone (zoom with two fingers).
☰ Tabs	Opens a clear list of every tab with the current one marked, a ✕ Close button and a Back button to the page you were on.
Mood meters	On other tabs, the Futures and Picks mood meters scroll away with the page.
IBKR and Webull boards	Tiles like the full board, two across, instead of only price and percent.
Bottom bar	Sits at the very bottom on every page, including System Check, where Menu now shows.
Sideways scrolling	If a page slides left or right, it eases back to center when you let go.
Phone page	Your Holdings moved below the Order Ticket. The Moves, Ticket, My Trading chips fit on one line.
Night Desk	The section buttons (Trade, ETF Edge, News and Feeds…) sit in an even grid.

New files: web/phone-polish.js, web/board-phone.css (added to board.css). Changed: looks.js, index.html, board.js, legend.js. Backup: backup-jb-phone/.

Algo safety (October 11)

Added on October 11 with the safety install (jb-ready), before any live money. The algo tabs' **? Legend** explains the same.

Where	What it does
Buying power check	Live only: before every buy, the algo compares the order with the buying power your broker reports (read every minute). Not enough: the buy is skipped and the algo tab says why. Selling is never blocked.
No buy-back after a stop	After a stop-loss, take-profit or trailing stop, the algo waits until its buy signal has switched off once before it can buy again (paper, live and backtests alike).
Stop works while paused	The Stop button now closes the position even when the algo is paused (after 3 rejected orders or the speed limit). It still checks the broker first, so nothing is sold twice.
Exits retried	An exit order that did not fill is sent again a little later. It no longer pauses the algo while it still holds shares.
IBKR prices stopped	If the IBKR algo gets no prices for 5 minutes (regular hours; longer at night) it asks IBKR again and pushes your phone. Urgent when it holds a position, since the stop-loss needs prices. Link to IBKR lost (code 1100) = safe mode until it is back.
Backup stop at the broker	Live only: right after each buy, the algo also places a stop order at IBKR or Webull, 1% wider than its own stop-loss (10% if the strategy has none), so the shares stay protected if the Mac or the algo is down. The algo cancels it before it sells, so nothing is sold twice. Hours covered: see Backup stop hours under Review fixes (October 11). Set risk: broker_stop_pct in the strategy file to change it (0 = off).
Loss limit pauses algos	Daily loss limit on My trading now also pauses both algos when it is hit (turned on October 11).

Changed files: both algos' core_trader.py, paper_trader.py, backtest.py and strategy.py, plus web/legend.js.

Review fixes (October 11)

Added on October 11 and 12 after two full code reviews (jb-reviewfix): every problem they found is fixed, mostly about selling twice, stops and the daily loss limit. Only the phone tiles and the notes on the algo tab look different. The **? Legend** explains the same.

Where	What it does
SELL ALL comes first	While SELL ALL is selling a position, that algo sends no sell of its own (no double sell). Pressing Stop during SELL ALL keeps it that way.
SELL ALL retry	If SELL ALL could not reach the broker when you pressed it (for example IB Gateway was restarting), the automatic follow-up a minute later presses it again. A stock bought after the press is still never sold by the follow-up.
Sells have no dollar cap	The order ticket's per-order limit (JB_TICKET_MAX, $5,000 to start) now applies to buys only. Board stops, Flip and scheduled sells work for any size; a sell is still never more than you hold.
Flip and the buy limit	Flip's buy half still counts as a buy for JB_TICKET_MAX. If the opposite fund would cost more than that, Flip says so before anything is sold (raise JB_TICKET_MAX to flip bigger positions).
Rejected order	A rejected order pauses the algo's BUYS for 15 minutes (3 in a row: until restart). Its stop-loss, exits and trailing stop keep working and retry after 1 minute.
Speed limit counts real orders	The algo's order speed limit (3 a minute, 30 a day) now counts only orders it really sends. Buys it skipped while waiting after a rejection no longer add up to a false "too many orders" pause.
LIVE trader watched	While ./jb live runs, your phone gets a loud push if its Terminal window is gone (closed, crash, Mac restart) or it stops checking in. Always on.

Where	What it does
LIVE trader crash	If the live trader stops with an error while its Terminal window stays open, your phone also gets one loud push (it no longer watches the stop-loss). Ctrl+C sends no push.
Backup stop hours	Each backup stop line on the algo tab says which hours it covers. IBKR: 4 am - 8 pm (IBKR takes no stop orders overnight). Webull: asked for all sessions, falling back to 9:30 - 4 if Webull refuses; check the order in the Webull app. Overnight, only the algo's own stop-loss protects the shares, so keep ./jb live running.
Backup stop no answer	If the broker gives no answer about the backup stop, the algo's exit waits (selling then could sell twice) and your phone is told once. The stop itself still protects the shares.
Backup stop gone	If the broker gives no answer about the backup stop but its own list of open orders shows no such stop (checked twice, 10 seconds apart) while you still hold all the shares, the stop is gone, so the algo's exit sells instead of waiting forever.
Overnight prices	If prices stop overnight, the algo asks again for the overnight prices too (IBKR overnight venue; Webull only if your plan has overnight).
Webull board overnight	If the Webull board gets no prices at all for 15 minutes overnight (when your Webull plan has overnight prices), it reconnects by itself, as it already did in the day. The board's own safety stops on Webull stocks need those prices.
Stops after a restart	A board safety stop that was selling when the board restarted goes back on, and sells only what you still hold.

Where	What it does
No stacked sells	Before a board stop, Flip or scheduled sell sends its sell, it cancels the board's own open sell orders for that stock (for example one sent just before a restart). If one can't be cancelled, nothing new is sent and your phone is told: this way the board never has more shares up for sale than you hold.
IBKR stop cancelled first	If you placed a real stop at IBKR from the protection check (Place at IBKR) and the board's own safety stop sells first (overnight), the board now cancels that IBKR stop before it sells. Before, the IBKR stop stayed open and could sell the same shares again at 4 am, leaving you short.
Monday numbers	The board now keeps 100 hours of prices, so Friday's close is still there on Monday: vs-close numbers, morning cards and the loss limit work on Mondays and after holidays. The last close knows holidays and 1 pm early closes.
Loss limit can lag	The daily loss limit also counts positions in symbols that are on no board (Yahoo price). Fills arrive every 5 min (IBKR) or 10 min (Webull), so the total can lag by that much.
Loss limit and futures	Only stocks and ETFs on no board are priced from Yahoo. A future (such as CL=F) counts toward the daily loss limit only while it is on the IBKR board: Yahoo would have looked up a different company (CLF) and shown a huge false loss.
Password needed to trade	With no JB_PASSWORD set, the board refuses orders, SELL ALL and algo changes (anyone on your Wi-Fi could otherwise press them). Requests from hidden frames on other websites are refused.
Login works again	The October 11 hidden-frame protection also refused the board's own login form (browsers send it the same way), so a new phone or browser could not log in. The login page is allowed again; orders, SELL ALL and algo changes from hidden frames are still refused.

Where	What it does
Phone tiles fit	On a phone, the long chips on the IBKR and Webull board tiles (SPREAD, AH PEAK) now wrap onto two lines inside the tile instead of being cut off and making the page scroll sideways.
Paper exits while paused	In paper mode too, a pause (speed limit, rejected orders) stops new buys only; the paper stop-loss and exits still run, the same as live.
Your own broker orders	When a board stop, Flip or scheduled sell sells shares, the board never cancels sell or stop orders YOU placed in the IBKR or Webull app, but the push now says if one is still open for that stock, so you can cancel it before it sells again.
Odd links hidden	News, press release, Truth Social and web-watch items only show a link when it is a normal web address (http or https).

Changed files: both algos' core_trader.py and paper_trader.py, gateway (algoctl, feeds, gateway, guardian, killswitch, orders, ticket, trading, watch, webwatch), the Webull bridge and scanner feed, the Webull live broker, web/legend.js, web/nd-feeds.js, web/plus.js and ./jb.

Use This Strategy (October 12)

Added on October 12 (jb-usestrat): switch the paper algo to a new strategy from the algo tab, with no Terminal. The algo tabs' **? Legend** explains the same.

Where	What it does
Use This Strategy	Next to Run backtest on each algo tab. The paper algo runs the strategy picked in the list from now on: the board changes that algo's line in settings.env and restarts it (about 30 seconds; the Strategy line updates). It asks first and says if you have not backtested that strategy yet.

Where	What it does
When it says no	While the paper algo holds a position (press Stop algo first, which closes it), while ./jb live runs, or when the file is not a valid strategy.
Paper only	It never starts or changes live trading: going live is still ./jb live with confirm_live: true.
Plain English to paper	Write the idea in Algo notes, tap Copy for Claude: Build It, paste Claude's file under Paste the strategy Claude wrote, press Run backtest, then Use This Strategy.

Delta Hedging (October 12)

The board measures how much your holdings across both brokers move with the Nasdaq-100 and S&P 500, and shows how to bring that close to flat. It uses only the board's own prices and free daily closes, and nothing here places an order by itself.

Where	What it does
Net Exposure	Portfolio tab (above Positions), My Trading tab and the phone page. Dollars gained or lost per 1% move in the Nasdaq-100 and S&P 500: each holding's value x leverage (3x funds count 3, inverse funds negative) x how closely it follows the index. IBKR futures count at full size. Details lists every holding.
Hedge Calculator	Portfolio tab, under Positions, and the phone card. Fully automatic, no inputs: how many shares of an inverse fund (SQQQ, SPXU, SOXS and others) or MNQ / MES micro futures (IBKR only) bring your exposure to about flat, with cost, daily decay and how much smaller your swings would have been. Best Match is marked.
Already About Flat	Shown when your exposure is under $5 per 1% move: nothing is worth hedging.

Where	What it does
Copy To Ticket	Fills the board's order ticket with the hedge. Nothing is sent until you press Check order, then Send. Futures: Copy Order Text.
Hedge Partners	Portfolio tab. For each stock or fund you hold: its best hedge partner (inverse twin, sector inverse, index inverse or micro future), how well each one hedged daily and overnight on the board's record, and the shares or contracts for your position.
Hedge Ideas	My Trading tab. Research, not advice: ranked pair ideas backtested on the board's own record for after-hours, overnight and pre-market, refreshed every 6 hours (Run Now to refresh). Send To Run Backtest writes the strategy into an algo and starts its backtest.
With Hedge	Algo tab backtests of a strategy with a hedge show the result alone and With Hedge.
Auto-Hedge	Each algo tab, in the Paper / live trading card. Default OFF. One click hedges that algo's position automatically: opens the hedge after the buy fills, shrinks it on a part exit, closes it with the position. Pick the fund (Auto picks the best) and the ratio (10-150%).
Auto-Hedge live	Its own switch, needs the typed words HEDGE LIVE and a live run going. A new live run always starts with live hedging off.
Hedge safety	Stop, the daily loss limit and SELL ALL close the hedge and open nothing new. The hedge's gains and losses count toward the daily loss limit. A rejected hedge waits 15 minutes. A hedge order left open from an earlier run stops live trading from starting, on both brokers.
Estimates	Futures margin (5-10%) and fund decay are estimates. Hedge Ideas use a few weeks of the board's record. Not yet tried on the real brokers.

Go Live and Webull Upgrades (October 12)

Live trading can now start and stop from each algo tab, with no Terminal. The Webull algo has the same safety checks as IBKR wherever Webull allows, and it turns overnight on by itself once you buy Webull's overnight data.

Where	What it does
Live Trading (Real Money) card	Each algo tab. Shows the strategy, the go-live checklist (fix any !! lines first), how the last live run ended and the live job's latest log lines.
Go Live	Type I_UNDERSTAND_THE_RISK, press Go Live and confirm. The paper algo stops and live trading runs in the background with the paper algo's strategy. Refused if the strategy lacks confirm_live: true or an order cap, or a live run is already going.
Stop Live	Stops live trading cleanly: working orders are cancelled, the position stays, and the paper algo starts again. A Terminal ./jb live run is still stopped with Control+C there.
LIVE banner	Red banner on the algo tab and phone page and a LIVE badge on the tab buttons while live trading runs. ./jb status lists it.
Start After A Mac Restart	Off by default: after a power cut or restart, live trading stays off until you press Go Live again. If live trading stops on its own, the phone gets a loud push.
./jb live	Still works exactly as before.
Webull: prices stopped	Same wait times as IBKR (5 min regular hours, 15 min pre and after hours, 30 min overnight), reconnects, and one phone push per outage.
Webull: connection lost	When Webull's link drops or another login takes the market data, safe mode holds orders until prices and the position check out again.

Where	What it does
Webull: short sales	Live only on the margin account, after Webull's own 1-share preview accepts it. Never in the overnight session.
Webull: leftover hedge	A hedge order left open from an earlier run stops live trading from starting, like IBKR.
Webull overnight	Off until you buy Webull's overnight data. Then the board, scanner and algo switch it on by themselves within 3 hours, or by 7:45 pm ET at the latest, with no restart. The Webull algo tab shows Overnight Prices (Webull): On / Off.
Webull overnight gap	The overnight gap strategy is now in the Webull algo's list too, with backtests on Webull data.
Not yet tried for real	Go Live on the Mac's launchd, Webull short sales and overnight stop orders, and real broker acceptance of hedge orders. Try 1 share first.

iPhone Widgets Update (October 14)

The Lock Screen widget now shows when it last got data from the board, and both widgets make a board outage impossible to miss.

Where	What it does
Lock Screen: IBKR Up · 8:26 PM · Webull Up	The middle time is when the widget last got fresh data from the board.
Lock Screen: Board Down	When the board can't be reached: a large ⚠️ Board Down, Can't Reach The Board, and the time the phone tried. The one-line style shows ⚠️ Board Down and the time.
Home Screen: Board Down	When the board can't be reached, the whole widget turns red: ⚠️ Board Down, Can't Reach The Board, Is Tailscale On? and Tried At 8:26 PM.

Where	What it does
Home Screen: chart backup prices	If the Yahoo prices are missing, the mini futures chart uses the board's own live futures prices from IBKR instead. Same size and layout as before.
Chart Waiting For Prices	Shown in place of the chart when there is nothing to draw yet. Harmless; it fills in by itself.
After updates	Copy the script into Scriptable again (Night desk, iPhone widget, Show my setup steps).

Weekly IBKR Login (October 14)

IBKR needs a full login with your password about once a week; the board now does it every Sunday at 5 PM ET, so it never runs out at a random time.

Where	What it does
Sunday 5 PM ET	IB Gateway restarts, the auto-login types your password, and your phone gets a Weekly IBKR Login push. Tap Approve in IBKR Mobile. This is an hour before futures open.
IBKR Login Expired	If IB Gateway ever says its security tokens expired, the board starts a fresh login at once and pushes you the same way. At most once every 30 minutes.
Don't log in by hand	While the auto-login is on, a second login (by hand in IB Gateway, or IBKR on the web) knocks the first one off in a loop. Let the auto-login do it and just tap Approve.
IB Gateway window	Normally minimized in the Dock, so it may not look open. ./jb status shows ibgateway running.
Historical data farm	A broken historical data farm message in IB Gateway is on IBKR's side and harmless; the board doesn't use it for live prices.

New file: web/algo-use.js. Changed: gateway/algoctl.py, web/legend.js, web/algo.html (one script line).

Emergency plan: the board is down and I hold a position

Protect the position first, then fix the board. The broker apps work without the Mac, and board safety stops (MAC ONLY) do not.

1. Open the **IBKR** or **Webull** app on your phone. Check what you hold and any open orders.
2. Close the position, or place a stop at the broker. Outside regular hours use a limit order: IBKR's overnight venue or Webull's night session.
3. If the app will not work, call the broker:
   - IBKR: 1 (877) 442-2757 toll-free, or 1 (312) 542-6901. Sunday 1 pm ET to Friday 7:30 pm ET.
   - Webull: 1 (888) 828-0618. Monday to Friday, 8 am to 5 pm ET only, so use the app overnight.
4. Then restart the board. Phone page: the Restart buttons on the red line, or the Health tab. At the Mac: check the charger, then `cd ~/josephs-board && ./jb restart` and `./jb status`.
5. If the home internet is down, turn on Personal Hotspot on your iPhone at home; the board switches to it when Backup internet is set up.

Monthly drill (5 minutes, first Sunday): open both broker apps and sign in, find the close button for a position, and check the phone numbers still work from the broker websites. Numbers checked Oct 9, 2026 on IBKR customer service and Webull help.